

Foundations of Quantum Computing

Portfolio Assessment

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Week 2: Properties of the Operator H

Consider the operator

$$H = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}.$$

(a) To show that H acts as a reflection on 2D vectors with real-valued coefficients, I compute H^2 directly:

$$\begin{aligned} H^2 &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 \cdot 1 + 1 \cdot 1 & 1 \cdot 1 + 1 \cdot (-1) \\ 1 \cdot 1 + (-1) \cdot 1 & 1 \cdot 1 + (-1) \cdot (-1) \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I. \end{aligned}$$

Since $H^2 = I$ and $H \neq I$, applying H twice returns every real-valued 2D vector to itself. A non-identity linear map on $\mathbb{R}^2$ that satisfies $M^2 = I$ is, by definition, a reflection. Therefore H is a reflection.

(b) Applying H to $|0\rangle$ gives

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle),$$

which makes an angle of 45° with the $|0\rangle$ axis in the real plane. For a reflection, the axis of reflection bisects the angle between any vector and its image. Since $|0\rangle$ (at 0°) is mapped to $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ (at 45°), the axis lies exactly halfway between them, at

$$\frac{0^\circ + 45^\circ}{2} = 22.5^\circ = \frac{\pi}{8}$$

from the $|0\rangle$ axis.

(c) I identify the eigenvalues and eigenvectors using the results of parts (a) and (b).

Eigenvalues. Since $H^2 = I$, every eigenvalue λ satisfies $\lambda^2 = 1$, so the eigenvalues are $+1$ and -1 .

Eigenvectors. A reflection preserves vectors lying along its axis (eigenvalue $+1$) and negates vectors perpendicular to it (eigenvalue -1). From part (b), the axis is at angle $\pi/8$, so the two orthonormal eigenvectors are:

$$\begin{aligned} |e_+\rangle &= \cos \frac{\pi}{8} |0\rangle + \sin \frac{\pi}{8} |1\rangle, & \text{eigenvalue } +1, \\ |e_-\rangle &= -\sin \frac{\pi}{8} |0\rangle + \cos \frac{\pi}{8} |1\rangle, & \text{eigenvalue } -1. \end{aligned}$$

These are orthogonal ($\langle e_+ | e_- \rangle = -\cos \frac{\pi}{8} \sin \frac{\pi}{8} + \sin \frac{\pi}{8} \cos \frac{\pi}{8} = 0$) and each is unit-normalised by $\cos^2 \frac{\pi}{8} + \sin^2 \frac{\pi}{8} = 1$.

Eigendecomposition. Writing H as a sum of rank-1 outer products weighted by eigenvalues:

$$H = |e_+\rangle\langle e_+| - |e_-\rangle\langle e_-|$$

where $|e_+\rangle = \cos \frac{\pi}{8} |0\rangle + \sin \frac{\pi}{8} |1\rangle$ and $|e_-\rangle = -\sin \frac{\pi}{8} |0\rangle + \cos \frac{\pi}{8} |1\rangle$. Thus H decomposes into a projection onto its $+1$ eigenspace minus a projection onto its -1 eigenspace, which is the spectral representation of a reflection.

Week 3: The SWAP Gate

Consider the operator

$$\text{SWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

which acts on two-qubit states as $\text{SWAP} |a, b\rangle = |b, a\rangle$.

(a) I determine the inverse of SWAP by considering its action on product states. Applying SWAP twice gives

$$\text{SWAP}^2 |a, b\rangle = \text{SWAP} |b, a\rangle = |a, b\rangle,$$

so $\text{SWAP}^2 = I$, which means $\text{SWAP}^{-1} = \text{SWAP}$. In other words, the SWAP gate is its own inverse.

Since $\text{SWAP}^2 = I$, every eigenvalue λ satisfies $\lambda^2 = 1$, and therefore the eigenvalues of SWAP are $+1$ and -1 .

(b) I seek a normalised state $|\psi\rangle = u_{00}|00\rangle + u_{01}|01\rangle + u_{10}|10\rangle + u_{11}|11\rangle$ satisfying $\text{SWAP} |\psi\rangle = -|\psi\rangle$.

Applying SWAP term by term:

$$\text{SWAP} |\psi\rangle = u_{00}|00\rangle + u_{01}|10\rangle + u_{10}|01\rangle + u_{11}|11\rangle.$$

Setting this equal to $-|\psi\rangle$ gives the constraints: $u_{00} = -u_{00}$ (so $u_{00} = 0$), $u_{11} = -u_{11}$ (so $u_{11} = 0$), and $u_{01} = -u_{10}$.

Choosing $u_{01} = \frac{1}{\sqrt{2}}$ (positive real) and $u_{10} = -\frac{1}{\sqrt{2}}$, and normalising, the (-1) -eigenvector is:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

This state has $u_{00} = u_{11} = 0$ and $u_{01} = -u_{10} = \frac{1}{\sqrt{2}}$, which is exactly the antisymmetric Bell-like state $|\Psi^-\rangle$.

(c) Since SWAP is Hermitian with eigenvalues ± 1 , it can be treated as an observable. To find the probability of the outcome -1 , I decompose $|01\rangle$ in the SWAP eigenbasis.

The $+1$ eigenspace is spanned by the symmetric states: $|00\rangle$, $|11\rangle$, and $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$. The -1 eigenspace is spanned by $|\psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$.

I write $|01\rangle$ as a superposition of symmetric and antisymmetric parts:

$$|01\rangle = \frac{1}{\sqrt{2}} \cdot \frac{|01\rangle + |10\rangle}{\sqrt{2}} + \frac{1}{\sqrt{2}} \cdot \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

so $|01\rangle = \frac{1}{\sqrt{2}}|\psi^+\rangle + \frac{1}{\sqrt{2}}|\psi^-\rangle$ where $|\psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$.

The probability of the -1 outcome is the squared amplitude of the $|\psi^-\rangle$ component:

$$P(-1) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}.$$

After obtaining the outcome -1 , the state is projected onto the -1 eigenspace and normalised. The post-measurement state is therefore:

$$P(-1) = \frac{1}{2}, \quad \text{post-measurement state: } \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

Thus, a SWAP measurement on $|01\rangle$ yields outcome -1 with probability $\frac{1}{2}$, leaving the system in the antisymmetric superposition $|\Psi^-\rangle$.

(d) The coherently controlled-SWAP (CSWAP) gate acts on three qubits: a control qubit c and two target qubits. It applies SWAP to the targets when $c = |1\rangle$ and acts as the identity otherwise, in direct analogy with the CU construction.

As an 8×8 matrix (ordering $|000\rangle, |001\rangle, |010\rangle, |011\rangle, |100\rangle, |101\rangle, |110\rangle, |111\rangle$):

$$\text{CSWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

In the more compact algebraic (projector) notation:

$$\text{CSWAP} = |0\rangle\langle 0| \otimes I_4 + |1\rangle\langle 1| \otimes \text{SWAP}$$

where I_4 is the 4×4 identity acting on the two target qubits. This mirrors the general CU form $|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U$, here with $U = \text{SWAP}$.

Week 4: Quantum Teleportation

Suppose Alice holds qubits $\mathbf{p}$ and $\mathbf{a}$, Bob holds qubits $\mathbf{b}$ and $\mathbf{q}$, with $\mathbf{a}$ and $\mathbf{b}$ pre-shared in the Bell state $|\Phi^+\rangle_{ab} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)_{ab}$, and $\mathbf{p}, \mathbf{q}$ in the unknown joint state $|\psi\rangle = u_{00}|00\rangle + u_{01}|01\rangle + u_{10}|10\rangle + u_{11}|11\rangle \in \mathbb{C}^4$.

(a) The four-qubit state, written with qubits in the order $\mathbf{p}, \mathbf{a}, \mathbf{b}, \mathbf{q}$, is the tensor product $|\psi\rangle_{pq} \otimes |\Phi^+\rangle_{ab}$ reordered to interleave the $\mathbf{a}$ and $\mathbf{b}$ registers:

$$\begin{aligned} |\Psi\rangle_{pabq} &= \frac{1}{\sqrt{2}} \sum_{i,j \in \{0,1\}} u_{ij} |i\rangle_p (|00\rangle + |11\rangle)_{ab} |j\rangle_q \\ &= \frac{1}{\sqrt{2}} \sum_{i,j} u_{ij} (|i, 0, 0, j\rangle + |i, 1, 1, j\rangle) \\ &= \frac{1}{\sqrt{2}} \left[u_{00} |0000\rangle + u_{01} |0001\rangle + u_{10} |1000\rangle + u_{11} |1001\rangle \right. \\ &\quad \left. + u_{00} |0110\rangle + u_{01} |0111\rangle + u_{10} |1110\rangle + u_{11} |1111\rangle \right] \end{aligned}$$

This is the starting state of the teleportation protocol; qubit ordering is $\mathbf{p}, \mathbf{a}, \mathbf{b}, \mathbf{q}$ throughout.

(b) I trace each operation in sequence, presenting all possible measurement outcomes together.

Operation 1: $CX_{\mathbf{p},\mathbf{a}}$ (Alice applies CNOT, control $\mathbf{p}$, target $\mathbf{a}$).

The gate acts as $|i, k, b, j\rangle \mapsto |i, i \oplus k, b, j\rangle$. Applying to each term of $|\Psi\rangle$:

$$|\Psi_1\rangle = \frac{1}{\sqrt{2}} \sum_{i,j} u_{ij} (|i, i, 0, j\rangle + |i, \bar{i}, 1, j\rangle)$$

where $\bar{i} = 1 \oplus i$. Grouping by the value of qubit $\mathbf{a}$:

$$|\Psi_1\rangle = \frac{1}{\sqrt{2}} \left[\underbrace{u_{00} |0, 0, 0, 0\rangle + u_{01} |0, 0, 0, 1\rangle + u_{10} |1, 0, 1, 0\rangle + u_{11} |1, 0, 1, 1\rangle}_{\mathbf{a}=0} + \underbrace{u_{10} |1, 1, 0, 0\rangle + u_{11} |1, 1, 0, 1\rangle + u_{00} |0, 1, 1, 0\rangle + u_{01} |0, 1, 1, 1\rangle}_{\mathbf{a}=1} \right]$$

Operation 2: $M_z \mathbf{a} \rightarrow r$ (Alice measures $\mathbf{a}$ in the computational basis).

Both outcomes $r \in \{0, 1\}$ occur with probability $\frac{1}{2}$.

If $r = 0$: the $\mathbf{a} = 0$ branch survives and normalises to

$$u_{00} |0, 0, 0, 0\rangle + u_{01} |0, 0, 0, 1\rangle + u_{10} |1, 0, 1, 0\rangle + u_{11} |1, 0, 1, 1\rangle.$$

If $r = 1$: the $\mathbf{a} = 1$ branch survives and normalises to

$$u_{10} |1, 1, 0, 0\rangle + u_{11} |1, 1, 0, 1\rangle + u_{00} |0, 1, 1, 0\rangle + u_{01} |0, 1, 1, 1\rangle.$$

Operation 3: $CX_{\mathbf{b},\mathbf{q}}$ (Bob applies CNOT, control $\mathbf{b}$, target $\mathbf{q}$).

The gate flips $\mathbf{q}$ when $\mathbf{b} = 1$. Applying separately to each branch:

Branch $r = 0$ (the $\mathbf{b}$ values are 0, 0, 1, 1):

$$u_{00} |0, 0, 0, 0\rangle + u_{01} |0, 0, 0, 1\rangle + u_{10} |1, 0, 1, 1\rangle + u_{11} |1, 0, 1, 0\rangle$$

Branch $r = 1$ (the $\mathbf{b}$ values are 0, 0, 1, 1):

$$u_{10} |1, 1, 0, 0\rangle + u_{11} |1, 1, 0, 1\rangle + u_{00} |0, 1, 1, 1\rangle + u_{01} |0, 1, 1, 0\rangle$$

Operation 4: $M_x \mathbf{b} \rightarrow s$ (Bob measures $\mathbf{b}$ in the X -basis).

Writing $|0\rangle_{\mathbf{b}} = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$ and $|1\rangle_{\mathbf{b}} = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$, expanding each branch yields four equiprobable outcomes (each with overall probability $\frac{1}{4}$). The normalised state of $\mathbf{p}, \mathbf{q}$ for each combined outcome is:

r	s	State of $\mathbf{p}, \mathbf{q}$ after both measurements
0	0	$u_{00} 00\rangle + u_{01} 01\rangle + u_{10} 11\rangle + u_{11} 10\rangle$
0	1	$u_{00} 00\rangle + u_{01} 01\rangle - u_{10} 11\rangle - u_{11} 10\rangle$
1	0	$u_{01} 00\rangle + u_{00} 01\rangle + u_{10} 10\rangle + u_{11} 11\rangle$
1	1	$-u_{01} 00\rangle - u_{00} 01\rangle + u_{10} 10\rangle + u_{11} 11\rangle$

Thus Alice's measurement r and Bob's measurement s together determine which of these four states the qubits $\mathbf{p}, \mathbf{q}$ are left in.

(c) Applying $CX_{\mathbf{p}, \mathbf{q}}$ to $|\psi\rangle$ (control $\mathbf{p}$, target $\mathbf{q}$) gives the target state:

$$|\psi'\rangle = CX |\psi\rangle = u_{00} |00\rangle + u_{01} |01\rangle + u_{10} |11\rangle + u_{11} |10\rangle.$$

Comparing this with the table above, the required local corrections are:

- $r = 0, s = 0$: no operation needed — the state is already $|\psi'\rangle$.
- $r = 0, s = 1$: Alice applies $Z_{\mathbf{p}}$ (negates the $|1\rangle_{\mathbf{p}}$ amplitude).
- $r = 1, s = 0$: Bob applies $X_{\mathbf{q}}$ (swaps $|0\rangle_{\mathbf{q}} \leftrightarrow |1\rangle_{\mathbf{q}}$).
- $r = 1, s = 1$: Alice applies $Z_{\mathbf{p}}$ and Bob applies $X_{\mathbf{q}}$.

To apply these corrections, Alice must send r (1 classical bit) to Bob, and Bob must send s (1 classical bit) to Alice.

(d) The protocol achieves something remarkable: Alice and Bob together apply the entangling gate CX to a two-qubit state $|\psi\rangle$ they do not know, even though qubit $\mathbf{p}$ is held by Alice and qubit $\mathbf{q}$ is held by Bob with no direct quantum channel between them. The entangled resource $|\Phi^+\rangle$ plays the role of a quantum “wire”: by consuming it together with two classical bits of communication, the parties can perform a non-local gate as if they were co-located. This is an instance of *gate teleportation*, and it demonstrates that shared entanglement can substitute for direct quantum interaction — a central insight for distributed quantum computing and quantum networks.

Week 5: Grover's Algorithm

Let $f : \{0, 1\}^2 \rightarrow \{0, 1\}$ satisfy $f(x) = 1$ iff $x = 01$, with oracle $\mathbf{0_f} |x_1 x_2, t\rangle = |x_1 x_2, t \oplus f(x_1 x_2)\rangle$. I trace the following procedure step by step, tracking the quantum state at each stage.

Initialisation: $\mathbf{MxReset} \mathbf{q}$ and $\mathbf{MxReset} \mathbf{t}$, then $\mathbf{Z} \mathbf{t}$

The two-qubit register $\mathbf{q}$ is set to $|+, +\rangle$ and the auxiliary qubit $\mathbf{t}$ is set to $|+\rangle$; applying Z then gives $Z |+\rangle = |-\rangle$. The joint initial state is:

$$|\phi_0\rangle = |+, +\rangle_{\mathbf{q}} \otimes |-\rangle_{\mathbf{t}} = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)_{\mathbf{q}} \otimes |-\rangle_{\mathbf{t}}$$

Oracle query: $\mathbf{0_f} q[0], q[1], t$

For the target qubit in state $|-\rangle$, the oracle produces the phase-kickback effect:

$$\mathbf{0_f} |x\rangle |-\rangle = (-1)^{f(x)} |x\rangle |-\rangle.$$

Since $f(01) = 1$ and $f(x) = 0$ for all other x , the oracle negates only the $|01\rangle$ amplitude:

$$|\phi_1\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle + |11\rangle)_{\mathbf{q}} \otimes |-\rangle_{\mathbf{t}}$$

The target qubit $|-\rangle$ is unchanged; from here I focus on the $\mathbf{q}$ register.

Diffusion step: R_u q

The subroutine R_u reflects $\mathbf{q}$ about $|+, +\rangle$. I apply each gate in sequence.

Sub-step 1 — $H q$ (Hadamard on each qubit):

Using $H|0\rangle = |+\rangle$ and $H|1\rangle = |-\rangle$, I compute $(H \otimes H)$ applied to $\frac{1}{2}(|00\rangle - |01\rangle + |10\rangle + |11\rangle)$:

$$\begin{aligned} (H \otimes H) |\phi_1\rangle_{\mathbf{q}} &= \frac{1}{2}(|+\rangle|+\rangle - |+\rangle|-\rangle + |-\rangle|+\rangle + |-\rangle|-\rangle) \\ &= \frac{1}{4}[(1-1+1+1)|00\rangle + (1+1+1-1)|01\rangle + (1-1-1-1)|10\rangle + (1+1-1+1)|11\rangle] \\ &= \frac{1}{2}(|00\rangle + |01\rangle - |10\rangle + |11\rangle) \end{aligned}$$

Sub-step 2 — CZ $q[0], q[1]$ (phase -1 on $|11\rangle$):

$$\frac{1}{2}(|00\rangle + |01\rangle - |10\rangle - |11\rangle)$$

Sub-step 3 — $H q$ (Hadamard again):

$$\begin{aligned} (H \otimes H) \frac{1}{2}(|00\rangle + |01\rangle - |10\rangle - |11\rangle) &= \frac{1}{4}[(1+1-1-1)|00\rangle + (1-1-1+1)|01\rangle + (1+1+1+1)|10\rangle + (1-1-1-1)|11\rangle] \\ &= |10\rangle \end{aligned}$$

Sub-step 4 — $X q$ (bit-flip on each qubit):

$$X \otimes X |10\rangle = |01\rangle$$

After R_u, the register is in the pure state $|01\rangle_{\mathbf{q}}$.

Measurement: $M_z q \rightarrow a$

Measuring $\mathbf{q}$ in the computational basis yields outcome $a = 01$ with probability 1. The measurement is deterministic: the algorithm has found the unique marked input with certainty.

Significance

This demonstrates Grover's algorithm completing a successful search after a single Grover iterate. For a 2-bit search space ($N = 4$) with one marked item, the optimal number of iterations is $\lfloor (\pi/4)\sqrt{N} \rfloor = \lfloor \pi/2 \rfloor = 1$, which is exactly what was performed. Classically, finding the marked input requires up to $N = 4$ queries on average; the quantum algorithm finds it in $O(\sqrt{N}) = O(2)$ queries — here reduced to a single query that succeeds with probability 1. This quadratic speedup is the central achievement of Grover's algorithm.

Week 6: Shor's Algorithm

I factorise $N = 323$ using $a = 2$, with phase-estimation samples $x_1 = 7/24$ and $x_2 = 5/18$.

(a) The eigenvalues of U_a are $e^{2\pi i k/s}$ for $k = 0, \dots, s-1$, so each sample x_j is close to some k_j/s . I apply the continued-fraction algorithm to find the best rational approximation to each sample.

Sample $x_1 = 7/24$. The continued-fraction expansion is $7/24 = [0; 3, 2, 3]$, whose last convergent is $7/24$ itself ($\gcd(7, 24) = 1$), giving candidate denominator $s_1 = 24$.

Sample $x_2 = 5/18$. The expansion is $5/18 = [0; 3, 1, 1, 2]$, whose last convergent is $5/18$ itself ($\gcd(5, 18) = 1$), giving candidate denominator $s_2 = 18$.

For the same order s to be consistent with both samples, s must be divisible by both s_1 and s_2 :

$$s = \text{lcm}(s_1, s_2) = \text{lcm}(24, 18).$$

Since $24 = 2^3 \cdot 3$ and $18 = 2 \cdot 3^2$, we have $\text{lcm}(24, 18) = 2^3 \cdot 3^2 = 72$.

Verification. $7/24 = 21/72$ and $5/18 = 20/72$, confirming both samples are multiples of $1/72$. We can verify directly: $323 = 17 \times 19$, and the orders of 2 modulo 17 and 19 are 8 and 18 respectively, so $\text{ord}_{323}(2) = \text{lcm}(8, 18) = 72$ ✓

$$\boxed{s = 72}$$

(b) For the order s to be useful in factorising N , two conditions must hold:

1. s must be **even**, so that $a^{s/2}$ is a well-defined integer.
2. $a^{s/2} \not\equiv \pm 1 \pmod{N}$, so that the square root of unity $a^{s/2} \pmod{N}$ is *non-trivial* (i.e. not the trivial roots ± 1), which guarantees that $\gcd(a^{s/2} \pm 1, N)$ yields a non-trivial factor.

Here $s = 72$ is even, and we will confirm that $2^{36} \not\equiv \pm 1 \pmod{323}$ in part (c).

(c) I set $u = a^{s/2} \pmod{N} = 2^{36} \pmod{323}$ and compute this via the Chinese Remainder Theorem, since $323 = 17 \times 19$.

Step 1 — $2^{36} \pmod{17}$. The order of 2 modulo 17 is 8 (since $2^4 \equiv 16 \equiv -1$ and $2^8 \equiv 1 \pmod{17}$). Writing $36 = 8 \times 4 + 4$:

$$2^{36} \equiv 2^4 \equiv 16 \equiv -1 \pmod{17}.$$

Step 2 — $2^{36} \pmod{19}$. The order of 2 modulo 19 is 18 (since $2^9 \equiv 512 \equiv 18 \equiv -1$ and $2^{18} \equiv 1 \pmod{19}$). Writing $36 = 18 \times 2$:

$$2^{36} \equiv 1 \pmod{19}.$$

Step 3 — *Combine via CRT.* I seek u with $u \equiv -1 \equiv 16 \pmod{17}$ and $u \equiv 1 \pmod{19}$. Writing $u = 19k + 1$ and substituting: $19k \equiv 15 \pmod{17}$. Since $19 \equiv 2 \pmod{17}$ and $2^{-1} \equiv 9 \pmod{17}$, we get $k \equiv 9 \times 15 = 135 \equiv -1 \equiv 16 \pmod{17}$, so $k = 17m - 1$ and $u = 19(17m - 1) + 1 = 323m - 18$. Taking $m = 1$ to obtain $1 < u < 323$:

$$u = 305.$$

Verification. $305 \equiv 16 \equiv -1 \pmod{17}$ and $305 = 16 \times 19 + 1 \equiv 1 \pmod{19}$ ✓
 $305^2 = (323 - 18)^2 \equiv 18^2 = 324 = 323 + 1 \equiv 1 \pmod{323}$ ✓ $305 \not\equiv \pm 1 \pmod{323}$ (since $305 \neq 1$ and $305 \neq 322$) ✓

$$\boxed{u = 305, \quad u^2 = 93025 = 288 \times 323 + 1 \equiv 1 \pmod{323}}$$

(d) Since $u^2 \equiv 1 \pmod{N}$, we have $N \mid (u^2 - 1) = (u - 1)(u + 1)$, but $N \nmid (u - 1)$ and $N \nmid (u + 1)$ (because $u \not\equiv \pm 1$). A common divisor of $(u \pm 1)$ and N must therefore be a non-trivial factor. I compute:

$$\begin{aligned} u - 1 = 304 &= 16 \times 19, & \gcd(304, 323) &= \gcd(304, 17 \times 19) = 19, \\ u + 1 = 306 &= 18 \times 17, & \gcd(306, 323) &= \gcd(306, 17 \times 19) = 17. \end{aligned}$$

(Alternatively, $\gcd(304, 323)$: Euclidean algorithm gives $323 = 1 \times 304 + 19$, $304 = 16 \times 19$, so $\gcd = 19$. Then $323/19 = 17$.)

The prime factorisation of N is therefore:

$$\boxed{323 = 17 \times 19}$$

Thus Shor's algorithm successfully factors 323 into the primes 17 and 19 using two phase-estimation samples, continued-fraction approximation, and a single modular-arithmetic computation.